$$Linear\ regresion\_OLS$$

$$OLS: Ordinary\ least\ square\ method$$

$Equation\ of\ simple\ linear\ regression:\ y = b_0 + b_1 * x_1$

$Equation\ of\ multiple\ linear\ regression:\ y = b_0 + b_1 * x_1 + b_2 * x_2 + \cdots + b_n * x_n$

$b_0:\ bias\ \ responsible\ for\ lift\ the\ line$

$b_1: slope\ \ \ responcible\ for\ line\ rotation$

$bo\ and\ b1\ \ are\ coefficients$

$y_{actual}\ \ \ vs\ \ \ y_{prediction}$

$error = (y_a - y_p)$

$error\ we\ will\ calculate\ observation\ wise:\ \ \ loss$

$Total\ error\ might\ become\ zero$

$squared\ error:\ (y_a - y_p)^2$

$$sum\ of\ squared\ error:\ \ SSE = \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} (\boldsymbol{y_{a_i}} - \boldsymbol{y_{p_i}})^{\boldsymbol{2}} = \boldsymbol{RSS}$$

$$mean\ square\ error = \frac{1}{N} * \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} (\boldsymbol{y_{a_i}} - \boldsymbol{y_{p_i}})^{\boldsymbol{2}}$$

$$\boldsymbol{root\ mean\ square\ error = RMSE = \sqrt{\frac{1}{N} * \sum_{i=1}^{n} (y_{a_i} - y_{p_i})^2}}$$

$\boldsymbol{Goal: Find\ the\ suitable\ coefficients\ , to\ minmize\ the\ cost\ function}$

$$cost\ function = j = \frac{1}{N} * \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} (\boldsymbol{y_{a_i}} - \boldsymbol{y_{p_i}})^{\boldsymbol{2}}$$

$coeff\ are\ good === predictions\ are\ good === the\ error\ is\ less == good\ Line$

$\boldsymbol{How\ to\ find\ the\ coefficients\ or\ parameters:\ \ b_o, b_1, b_2, \ldots.}$

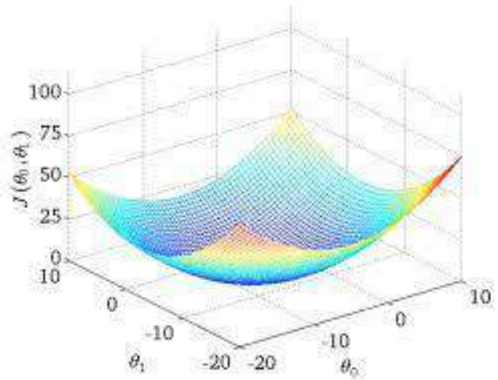
$\boldsymbol{coefficients\ are\ good}$

$\boldsymbol{it\ affects\ the\ cost\ function, which\ means\ \ your\ \ J\ will\ be\ minimized}$

$$\boldsymbol{y_{p_i} = b_o + b_1 * x_1}$$

$$J = \frac{1}{N} * \sum_{i=1}^{n} \left(y_{a_i} - y_{p_i}\right)^2$$

$$J = \frac{1}{N} * \sum_{i=1}^{n} \left(y_{a_i} - b_o - b_1 * x_1\right)^2 ===== \textbf{how the graph should be (parabola)}$$



How to find coefficients:

- *OLS: Ordinary least square (Linear regression and Logistic will use mainely)*
- *Maximum likely hood (Logistic regression mainely)*
- *Gradient descent(Deep learning can applicable for ML also)*

Oridinary Least Square method

- *we need to do minimize the cost function*
- *minimization means at that particular point , slope the equation will zero*
- *slope means i recall differentaition*
- *so differentatiation of equation will be zero*
- *but in above cost function we need to find bo and b1*
- *partial differentation*

$$\frac{\partial J}{\partial b_o} = 0 \quad and \quad \frac{\partial J}{\partial b_1} = 0$$

$$case - 1: \frac{\partial J}{\partial b_0} = 0$$

$$J = \frac{1}{N} * \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x})^{\boldsymbol{2}}$$

$$\frac{\partial J}{\partial b_0} = \frac{\partial \frac{1}{N} * \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x})^{\boldsymbol{2}}}{\partial b_0} = \frac{1}{N} * \frac{\partial \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x})^{\boldsymbol{2}}}{\partial b_0}$$

$$\frac{2}{N} * \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x})[\frac{\partial y_a}{\partial b_0} - \frac{\partial b_o}{\partial b_0} - \frac{\partial b_1 x}{\partial b_0}]$$

$$\frac{\partial J}{\partial b_0} = \frac{1}{n} * 2 \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i\right) [0 - 1 - 0]$$

$$\frac{\partial J}{\partial b_0} = \frac{1}{n} * 2 \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i\right) [-1]$$

$$\frac{\partial J}{\partial b_0} = \frac{-2}{n} \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i\right)$$

$$\frac{\partial J}{\partial b_0} = 0$$

$$\frac{-2}{n} \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i\right) = 0$$

$$\frac{-2}{n} \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i\right) = 0$$

$$\frac{1}{n} * \left(\sum_{i}^{n} y_{a_i} - \sum_{i}^{n} b_0 - \sum_{i}^{n} b_1 * x_i \right) = 0$$

$$here\ x_i\ is\ input\ data$$

$$here\ y_a\ is\ output\ data$$

$$\frac{1}{n} * \left(\sum_{i}^{n} y_{a_i} - b_o \sum_{i}^{n} 1 - b_1 \sum_{i}^{n} x_i \right) = 0$$

$$\sum_{i}^{n} b_0 = b_o \sum_{i}^{n} 1 = b_0 n$$

$$\sum_{i=1}^{n} 1 = 1 + 1 + 1 + \cdots n$$

$$\sum_{i}^{n} 1 = 1 + 1 + 1 + 1 + \cdots = n$$

$$\sum_{i}^{n} b_1 * x_i = b_1 \sum_{i}^{n} x_i$$

$$\frac{1}{n} * \left[\left(\sum_{i}^{n} y_{a_i} - b_0 * n - b_1 \sum_{i}^{n} x_i\right)\right] = 0$$

$$\frac{1}{n} * \sum_{i}^{n} y_{a_i} - \frac{1}{n} * b_o * n - \frac{1}{n} * b_1 \sum_{i}^{n} x_i = 0$$

$$\frac{1}{n} * \sum_{i}^{n} y_{a_i} = \frac{1}{n} * \left(y_{a_1} + y_{a_2} + \cdots . + y_{a_n}\right) = \bar{y}_a$$

$$\frac{1}{n} * \sum_{i}^{n} x_i = \bar{x}$$

$$(\overline{y_a} - b_0 - b_1 * \bar{x}) = 0$$

$$b_0 = \overline{y_a} - b_1 * \bar{x} ====== > Bias\ formuale$$

$$case - 1: \frac{\partial J}{\partial b_0} = 0$$

$$\frac{\partial J}{\partial b_0} = \frac{\partial \frac{1}{n} * \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x_i})^{\boldsymbol{2}}}{\partial b_0}$$

$$\frac{\partial J}{\partial b_0} = \frac{2}{n} * \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x_i})[\frac{\partial y_a}{\partial b_0} - \frac{\partial b_o}{\partial b_0} - \frac{\partial b_1 x_i}{\partial b_0}]$$

$$\frac{\partial y_a}{\partial b_0} = 0; \frac{\partial b_o}{\partial b_0} = 1; \frac{\partial b_1 x_i}{\partial b_0} = 0$$

$$\frac{\partial J}{\partial b_0} = \frac{1}{n} * 2 \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i \right) [0 - 1 - 0]$$

$$\frac{\partial J}{\partial b_0} = \frac{1}{n} * 2 \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i \right) [-1]$$

$$\frac{\partial J}{\partial b_0} = \frac{-2}{n} \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i \right)$$

$$\frac{-2}{n} \sum_{i=1}^{n} \left(y_{a_i} - b_0 - b_1 * x_i \right) = 0$$

$$-\frac{2}{n} \left(\sum_{i}^{n} y_{a_i} - \sum_{i}^{n} b_0 - \sum_{i}^{n} b_1 * x_i \right) = 0$$

b_0 *and* b_1 *are constants*

$$-\frac{2}{n} \left(\sum_{i}^{n} y_{a_i} - b_o \sum_{i}^{n} 1 - b_1 \sum_{i}^{n} x_i \right) = 0$$

$$\sum_{i}^{n} 1 = n$$

$$-\frac{2}{n} \left(\sum_{i}^{n} y_{a_i} - b_o n - b_1 \sum_{i}^{n} x_i \right) = 0$$

$$\frac{1}{n} * \sum_{i}^{n} y_{a_i} - \frac{1}{n} * b_o * n - \frac{1}{n} * b_1 \sum_{i}^{n} x_i = 0$$

$$\frac{1}{n} * \sum_{i}^{n} y_{a_i} = \frac{1}{n} * \left(y_{a_1} + y_{a_2} + \cdots . + y_{a_n} \right) = \bar{y}_a$$

$$\frac{1}{n} * \sum_{i}^{n} x_i = \bar{x}$$

$$(\overline{y_a} - b_0 - b_1 * \bar{x}) = 0$$

$$b_0 = \overline{y_a} - b_1 * \bar{x} ======> Bias\ formuale$$

$$case - 2: \frac{\partial J}{\partial b_1} = 0$$

$$\frac{\partial J}{\partial b_1} = \frac{\partial \frac{1}{n} * \sum_{i=1}^{n} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x_i})^2}{\partial b_1}$$

$$\frac{\partial J}{\partial b_1} = \frac{2}{n} * \sum_{i=1}^{n} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x_i})[\frac{\partial y_a}{\partial b_1} - \frac{\partial b_o}{\partial b_1} - \frac{\partial b_1 x_i}{\partial b_1}]$$

$$\frac{\partial y_a}{\partial b_1} = 0; \frac{\partial b_o}{\partial b_1} = 0; \frac{\partial b_1 x_i}{\partial b_1} = x_i$$

$$\frac{\partial J}{\partial b_1} = \frac{2}{n} * \sum_{i=1}^{n} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x_i})[0 - 0 - x_i]$$

$$\frac{\partial J}{\partial b_1} = \frac{2}{n} * \sum_{i=1}^{n} (\boldsymbol{y_a} - \boldsymbol{b_o} - \boldsymbol{b_1} * \boldsymbol{x_i})[-x_i]$$

$$b_o = \overline{y_a} - b_1 * \bar{x}$$

$$\frac{\partial J}{\partial b_1} = \frac{2}{n} * \sum_{i=1}^{n} (\boldsymbol{y_a} - (\overline{y_a} - b_1 * \bar{x}) - \boldsymbol{b_1} * \boldsymbol{x_i})[-x_i]$$

$$\frac{\partial J}{\partial b_1} = \frac{2}{n} * \sum_{i=1}^{n} (\boldsymbol{y_a} - \overline{y_a} + b_1 * \bar{x} - \boldsymbol{b_1} * \boldsymbol{x_i})[-x_i]$$

$$\frac{\partial J}{\partial b_1} = \frac{2}{n} * [\sum_{i=1}^{n}[(\mathbf{y_a} - \overline{y_a}) + b_1(\bar{x} - \mathbf{x_i})]] \, [-x_i]$$

$$\frac{\partial J}{\partial b_1} = \frac{2}{n} * \left[\sum_{i}^{n}(\boldsymbol{y_a} - \overline{y_a}) + \sum_{i}^{n} b_1(\bar{x} - \boldsymbol{x_i})\right] * [-x_i]$$

$$\frac{\partial J}{\partial b_1} = \left[\frac{2}{n} * \sum_{i}^{n}(\boldsymbol{y_a} - \overline{y_a}) + \frac{2}{n} * \sum_{i}^{n} b_1(\bar{x} - \boldsymbol{x_i})\right] * [-x_i]$$

$$\frac{\partial J}{\partial b_1} = \left[-\frac{2x_i}{n} * \sum_{i}^{n}(\boldsymbol{y_a} - \overline{y_a}) - \frac{2 * x_i}{n} * \sum_{i}^{n} b_1(\bar{x} - \boldsymbol{x_i})\right]$$

$$\frac{\partial J}{\partial b_1} = 0$$

$$-\frac{2}{n} * \sum_{i}^{n}(\boldsymbol{y_{a_i}} - \overline{y_{a_i}}) * x_i = \frac{2}{n} * \sum_{i}^{n} b_1(\bar{x} - \boldsymbol{x_i}) * x_i$$

$$-\sum_{i}^{n}(\boldsymbol{y_{a_i}} - \overline{y_{a_i}}) = \sum_{i}^{n} b_1(\bar{x} - \boldsymbol{x_i})$$

$$b_1 = \sum_{i}^{n}(\overline{y_a} - \boldsymbol{y_a}) \Big/ \sum_{i}^{n}(\bar{x} - \boldsymbol{x_i})$$

Ordinary least square method : OLS

$$b_0 = \overline{y_a} - b_1 * \bar{x}$$

$$b_1 = \frac{\sum_i^n(\overline{y_a} - \boldsymbol{y_{ai}})}{\sum_i^n(\bar{x} - \boldsymbol{x_i})}$$

$$b_1 = \frac{\sum_i^n(\overline{y_a} - \boldsymbol{y_{ai}}) * (\bar{x} - \boldsymbol{x_i})}{\sum_i^n(\bar{x} - \boldsymbol{x_i}) * (\bar{x} - \boldsymbol{x_i})}$$

$$b_1 = \frac{\sum_i^n(\overline{y_a} - \boldsymbol{y_{ai}}) * (\bar{x} - \boldsymbol{x_i})}{\sum_i^n(\bar{x} - \boldsymbol{x_i})^2}$$

$$b_1 = \frac{Cov(x,y)}{Var(x)}$$

$$Correlation\ (r) = \frac{Cov(x,y)}{\sigma_x * \sigma_y}$$

$$b_0 = \overline{y_a} - b_1 * \bar{x}$$

$$b_1 = \frac{\sum_i^n (\overline{y_a} - \boldsymbol{y_{ai}}) * (\bar{x} - \boldsymbol{x_i})}{\sum_i^n (\bar{x} - \boldsymbol{x_i})^2}$$

$$b_1 = \frac{Cov(x, y)}{Var(x)}$$